He found that it was due to a chemical produced by the mould and he named it as penicillin, which was the first antibiotic discovered by Alexander Fleming in 1926 **(Fig. 8.1)**. Penicillin is produced by the fungi *Penicillium notatum* and *Penicillium chrysogenum*. It is bactericidal (antibiotics that kill bacteria) in action and inhibits the synthesis of the bacterial cell wall.

Penicillin is also referred as the **"queen of drugs"** and its full potential as an effective antibiotic was established much later by **Earnest Chain** and **Howard Florey** when they treated the wounded soldiers in World War II with penicillin. **Fleming, Chain** and **Florey** were awarded the **Nobel prize** in 1945 for the discovery of penicillin.

Antibiosis is the property of antibiotics to kill microorganisms.
Broad-spectrum antibiotics act against a wide range of disease-causing bacteria.
Narrow-spectrum antibiotics are active against a selected group of bacterial types.

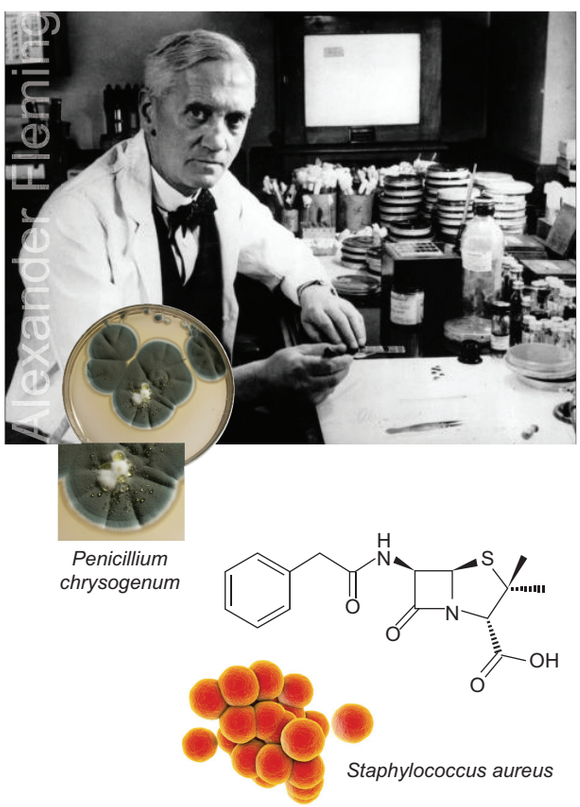


Fig. 8.1 Discovery of penicillin

DO YOU KNOW?

Hypersensitivity reaction is a major problem with the use of penicillin, resulting in nausea, vomiting, wheezing and ultimately cardiovascular collapse. To check the sensitivity reaction, doctors use a needle to prick the forearm of the patients to give a weak dose of penicillin. An itchy red region in the forearm is an indication that the patient is allergic to penicillin. This test is important before administration of penicillin to a patient.

Tetracycline is a broad spectrum bacteriostatic antibiotic (antibiotics that limit the growth of bacteria) that inhibits microbial protein synthesis. **Chlortetracycline** is the first antibiotic of this group, isolated from the cultures of *Streptomyces aureofaciens*. **Streptomycin** is a broad spectrum antibiotic isolated from the actinomycetes, *Streptomyces griseus*. It is bactericidal against both gram positive and gram negative bacteria, especially against *Mycobacterium tuberculosis*. Antibiotics, such as **erythromycin, chloromycetin, griseofulvin, neomycin, kenamycin, bacitracin,** etc., are also isolated as microbial products.

Antibiotic resistance

Antibiotic resistance occurs when bacteria develop the ability to defeat the drug designed to kill or inhibit their growth. It is one of the most acute threat to public health. Antibiotic resistance is accelerated by the misuse and over use of antibiotics, as well as poor infection prevention control. Antibiotics should be used only when prescribed by a certified health professional. When the bacteria become resistant, antibiotics cannot fight against them and the bacteria multiply. Narrow spectrum antibiotics are preferred over broad spectrum antibiotics. They effectively and accurately target specific pathogenic organisms and are less likely to cause resistance. "**Superbug**" is a term used to

XII Std Biology-Zoology Chapter-8 EM.indd 140 23/12/2022 12:45:09